Unbiasedness + exact sampling distributions

1 RECOGNISE IT

- Small finite population; asked to list all possible samples/statistics.
- Sampling rule and replacement status are specified.
- Asked whether an estimator is unbiased.

2 DECISION / METHOD

1. List all allowed samples under the stated sampling rule.
2. Compute the statistic for each sample.
3. Build its probability/frequency distribution.
4. Compare the mean of that distribution with the parameter.

3 FORMULA

`unbiased if E(estimator)=parameter`

$E(\bar{X})=\mu$

$E(p)=\pi$

4 FORMULA SHEET VS WHAT I KNOW

Gives: the inference table pairs $\bar{X}$ with μ and p with π.

Still know: exact enumeration, probabilities, and the expected-value definition of unbiasedness.

5 EXCEL ROUTE

For small enumeration tables:

- use rows for allowed samples
- `=AVERAGE(sample_cells)`
- average the full statistic column to check the centre.

6 TINY WORKED EXAMPLE

Population {2,4}; n=1 samples give $\bar{X}$ values 2 and 4 equally. Their mean is 3, equal to population $\mu=3$, so $\bar{X}$ is unbiased.

7 TOP EXAM TRAPS

- Unbiased does not mean every estimate equals the parameter.
- Ignoring replacement/order rules.
- Checking only one sample instead of E(estimator).

8 SAY ALOUD

Unbiased means the estimator is centred correctly over repeated samples, not that every sample is correct.

9 EXIT CHECK

What equality must hold for $\bar{X}$ to be an unbiased estimator of μ?

ONE-GLANCE VISUAL ROUTE

